From Predictive Agreement to Geometric Agreement: Stability of Fisher Connections and Semantic Transport in Neural Representations

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Abstract

A probabilistic decoder equips its predictively identifiable hidden-state quotient with the pullback Fisher metric and Amari-Chentsov tensor. We establish four statements about comparing this geometry across models. First, two embeddings with the same predictive image are isomorphic for every Amari α-connection, parallel transport, curvature, and every smooth field induced on that image. Second, uniform KL convergence alone does not imply geometric convergence: a uniformly interior and regular Bernoulli sequence has vanishing KL error while its pullback Fisher metric fails to converge at a fixed point. Third, on compact uniformly interior and nondegenerate families, C^2 predictive closeness controls the metric in C^1, every bounded-α connection, and parallel transport, while C^3 closeness controls curvature. With a uniform H^s bound, integrated KL controls transport at rate $O(\mathcal{K}^\beta)$ for every $\beta < (s-2-m/2)/(2s)$. Fourth, the conditional Fisher variance of a semantic predictive effect is exactly the irreducible squared error of all fields measurable from the predictive state. Separate orthogonal decompositions isolate this obstruction from connection mismatch and from cross-model field mismatch. At an affine softmax head, constant vector reuse is exactly exponential parallel transport; the results therefore support a controlled comparison of connections without selecting one in advance.

1 Introduction

Modern neural networks compute with vectors while producing probability distributions. For an autoregressive language model, a hidden state h entering the output head determines

$$F(h) = p(\cdot \mid h) \in \Delta_N^\circ, \tag{1}$$

where Δ_N° is the interior of the categorical simplex over a vocabulary of size N. The hidden coordinate remains a point of $\mathbb{R}^d$. Nevertheless, its predictively visible component parameterizes a statistical model and inherits the Fisher geometry of the decoder.

Exact equality of identifiable predictive images gives an isomorphism of their induced statistical structures. Approximate equality at the loss level is weaker: cross-entropy controls probability values on the data distribution, whereas Fisher metrics and affine connections depend on derivatives of the predictive map. Geometric equivalence also does not imply that a semantic operation is determined by the next-token distribution, because hidden states can contain decoder-null or semantically relevant fiber directions.

At a linear softmax head, constant hidden-vector reuse is exactly the affine operation selected by the flat exponential connection. Ordinary addition is therefore one connection-dependent rule rather than an alternative to geometry. The resulting question is which connection, if any, makes a specified semantic effect transferable across contexts and models.

The analysis proceeds through five results:

1. Exact predictive naturality: two identifiable parameterizations of the same predictive family preserve the Fisher metric, Amari-Chentsov tensor, every α-connection, parallel transport, and curvature up to reparameterization; every common smooth field on the shared predictive image pulls back equivariantly.

2. Separation of predictive and geometric convergence: a smooth, uniformly regular Bernoulli family converges uniformly in KL while its Fisher metric fails to converge at a fixed point.

3. Quantitative C^k stability: on compact uniformly interior and uniformly identifiable families, C^2 predictive closeness controls the defect between aligned α-connections and the commutator of their parallel transports, while C^3 closeness controls curvature.

4. A loss-to-geometry implication under explicit regularity: integrated KL convergence plus a uniform H^s bound implies C^2 convergence by Sobolev interpolation, and hence convergence of α-parallel transport with a quantitative rate.

5. Semantic conditional-variance decompositions: two orthogonal identities separately isolate failure to descend to the predictive quotient from connection mismatch and from cross-model conditional-mean-field mismatch.

These results define a concrete comparison protocol: choose a predictive quotient and a cross-model alignment, estimate semantic effects on that quotient, and compare exponential, Levi-Civita, mixture, and learned-α transport through commutation error and behavioral transfer.

2 Related work

Information geometry equips regular statistical models with the Fisher metric, the Amari-Chentsov cubic tensor, and dual affine connections [2, 3, 6]. Invariance under sufficient statistics and the uniqueness of the Fisher and Amari-Chentsov tensors are classical [5, 8]. The distinction between affine flatness and Levi-Civita flatness is standard; global exponential or mixture coordinates do not force zero Riemannian curvature [7].

Pullback geometry in learned latent spaces is also established. Arvanitidis et al. [4] pull Fisher-Rao geometry through probabilistic decoders. Zhang [18] explicitly connects a language model's hidden-to-predictive map, likelihood training, and pullback geometry. For softmax representations, Park et al. [12] develop an information-geometric account of probing and steering, while Wang and Zhao [15] pull the output Fisher metric through transformer layers and show large deviations from ambient Euclidean geometry. These results provide the decoder-side geometric setting used below.

Cross-model representation convergence has been studied from several directions. The Platonic Representation Hypothesis proposes convergence toward shared statistical structure [11]. Yu et al. [17] align distinct neural latent geometries with pullback metrics and relative geodesic representations. Hu et al. [10] model contextual transformations as vector fields and report shared displacement structure across language-model families. Path dependence and holonomy have also been proposed as representation diagnostics [14]. The stability analysis below asks when predictive agreement forces agreement of connection-dependent transport.

Continuity of Fisher information requires control of both distributions and their derivatives; Rezakhani et al. [13] prove such bounds in classical and quantum settings. Quantitative relaxations of sufficient statistics based on bi-Lipschitz Fisher metrics are studied by Yamaguchi and Nozawa [16]. Here a fixed comparison diffeomorphism aligns the predictive maps; their closeness is then propagated through every Amari α-connection to an explicit parallel-transport commutator bound. Sobolev regularity supplies the derivative control needed to derive this closeness from KL convergence.

3 Predictive statistical manifolds

3.1 The categorical simplex

Let

$$\Delta_N^\circ = \left\{ p \in \mathbb{R}^N : p_a > 0,\ \sum_{a=1}^N p_a = 1 \right\}. \tag{2}$$

For tangent vectors u, v, w satisfying $\sum_a u_a = \sum_a v_a = \sum_a w_a = 0$, define

$$g_p^\Delta(u,v) = \sum_{a=1}^N \frac{u_a v_a}{p_a}, \tag{3}$$

$$C_p^\Delta(u,v,w) = \sum_{a=1}^N \frac{u_a v_a w_a}{p_a^2}. \tag{4}$$

These are the Fisher metric and Amari-Chentsov tensor. A smooth predictive map

$$F : M \longrightarrow \Delta_N^\circ \tag{5}$$

induces

$$g_F = F^* g^\Delta, \qquad C_F = F^* C^\Delta. \tag{6}$$

When F is an immersion, g_F is a Riemannian metric. In local coordinates $x = (x^1, \ldots, x^m)$, writing $p_a = F_a(x)$,

$$(g_F)_{ij} = \sum_a \frac{\partial_i p_a \, \partial_j p_a}{p_a}, \tag{7}$$

$$(C_F)_{ijk} = \sum_a \frac{\partial_i p_a \, \partial_j p_a \, \partial_k p_a}{p_a^2}. \tag{8}$$

Raise the final index of C_F using g_F and define K_F by

$$g_F(K_F(X,Y), Z) = C_F(X,Y,Z). \tag{9}$$

We use the convention

$$\nabla_F^{(\alpha)} = \nabla_F^{\mathrm{LC}} - \frac{\alpha}{2} K_F. \tag{10}$$

Thus $\alpha = 0$ is Levi-Civita, while $\alpha = 1$ and $\alpha = -1$ are the exponential and mixture connections on a regular exponential family.

3.2 Predictive quotients

Neural predictive maps are commonly nonidentifiable. Define

$$h \sim_F h' \iff F(h) = F(h'). \tag{11}$$

When the fibers are regular, the identifiable state space is the quotient $M_F = M/{\sim_F}$. All geometric statements below concern this quotient or a chosen horizontal subbundle on which the pullback metric is nondegenerate.

For a linear softmax decoder,

$$F(h) = \mathrm{softmax}(Wh + b), \tag{12}$$

the null space is

$$\mathcal{N} = \{u : Wu \in \mathrm{span}\{\mathbf{1}\}\}. \tag{13}$$

The quotient $\mathbb{R}^d/\mathcal{N}$ embeds smoothly into the simplex, and before quotienting

$$G(h) = W^\top(\mathrm{diag}\, p_h - p_h p_h^\top)W, \qquad \ker G(h) = \mathcal{N}. \tag{14}$$

This is the rigorous sense in which the predictively identifiable component of a hidden space is a statistical manifold. A finite set of naturally occurring hidden states remains a point cloud unless a smooth intervention chart or interpolation model is supplied.

3.3 Ordinary addition selects a connection

For the affine softmax family

$$p_h(a) = \exp\big(w_a^\top h + b_a - A(h)\big), \tag{15}$$

natural hidden coordinates are exponential-affine. Therefore constant coordinate vectors are $\nabla^{(1)}$-parallel. More explicitly,

$$p_{h_r+h_q-h_p}(a) = \frac{p_{h_r}(a)p_{h_q}(a)/p_{h_p}(a)}{\sum_b p_{h_r}(b)p_{h_q}(b)/p_{h_p}(b)}. \tag{16}$$

Thus hidden vector arithmetic at the final affine head is exactly exponential transport, not the absence of geometry. Levi-Civita transport instead preserves Fisher lengths and angles, while mixture transport preserves the dual affine structure. Which connection best models a semantic operation is an empirical question.

4 Exact predictive naturality

Exact predictive equivalence determines the corresponding metric, tensors, connections, and transports.

Theorem 4.1 (Exact predictive naturality)**.** *Let $F_A : M_A \to \Delta_N^\circ$ and $F_B : M_B \to \Delta_N^\circ$ be smooth embeddings onto the same embedded predictive image P. Define*

$$\Phi = F_B^{-1} \circ F_A : M_A \longrightarrow M_B. \tag{17}$$

Then

$$\Phi^* g_B = g_A, \qquad \Phi^* C_B = C_A. \tag{18}$$

For every $\alpha \in \mathbb{R}$,

$$\mathrm{d}\Phi(\nabla^{(\alpha)}_{A}{}_{X} Y) = \nabla^{(\alpha)}_{B}{}_{\mathrm{d}\Phi X}(\mathrm{d}\Phi Y). \tag{19}$$

For every piecewise C^1 path γ,

$$\mathrm{d}\Phi_{\gamma(1)} P^{A,\alpha}_{\gamma} = P^{B,\alpha}_{\Phi\circ\gamma} \mathrm{d}\Phi_{\gamma(0)}. \tag{20}$$

Curvature tensors intertwine, and holonomies of corresponding loops are conjugate.

Proof. Since $F_A = F_B \circ \Phi$, functoriality of pullback gives

$$\Phi^* g_B = \Phi^* F_B^* g^{\Delta} = (F_B \circ \Phi)^* g^{\Delta} = F_A^* g^{\Delta} = g_A, \tag{21}$$

and identically for C. A Riemannian isometry preserves the Levi-Civita connection. Preservation of g and C implies preservation of K, so (10) gives (19). Applying $\mathrm{d}\Phi$ to an A-parallel field and using uniqueness of the linear transport ODE gives (20). Curvature naturality follows from its definition as the commutator of covariant derivatives, and loop holonomies are consequently conjugate. □

Corollary 4.2 (Intrinsic semantic fields)**.** *Let $S_t : P \to P$ be a smooth semantic flow with infinitesimal field*

$$Z_P(p) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} S_t(p). \tag{22}$$

Define model fields X_i by $\mathrm{d}F_i(X_i) = Z_P \circ F_i$. Then

$$\mathrm{d}\Phi(X_A) = X_B \circ \Phi. \tag{23}$$

The condition is substantive: a context operation need not descend to P. If two hidden states have the same prediction before an intervention but distinct predictive effects afterward, then no intrinsic field Z_P exists on the predictive quotient. Section 8 quantifies this obstruction.

5 Prediction convergence does not imply geometric convergence

Let $P(\cdot \mid c)$ be the data conditional and $q(\cdot \mid c)$ a model. Population cross-entropy decomposes as

$$\mathcal{L}(q) - H_P(Y \mid C) = \mathbb{E}_C D_{\mathrm{KL}}(P(\cdot \mid C) \| q(\cdot \mid C)). \tag{24}$$

The logarithmic score is strictly proper [9]. If models q_A, q_B have excess risks ϵ_A, ϵ_B, Pinsker's inequality and the triangle inequality yield

$$\mathbb{E}_C \| q_A(\cdot \mid C) - q_B(\cdot \mid C)\|_1 \le \sqrt{2\epsilon_A} + \sqrt{2\epsilon_B}. \tag{25}$$

This makes output agreement natural under realizability and global population-risk optimization. It provides no derivative control.

Proposition 5.1 (Regular KL-to-geometry counterexample)**.** *There exist smooth, uniformly interior, regular Bernoulli predictive families p_n, p_* on $[-1, 1]$ such that*

$$\sup_x D_{\mathrm{KL}}(p_*(x) \| p_n(x)) = O(n^{-2}), \tag{26}$$

but their pullback Fisher metrics do not converge even pointwise at $x = 0$.

Proof. Let $p_*(x) = \mathrm{Ber}(r(x))$ and $p_n(x) = \mathrm{Ber}(r_n(x))$, where

$$r(x) = \frac{1}{3} + \frac{x}{20}, \qquad r_n(x) = r(x) + \frac{1}{100n}\sin(nx). \tag{27}$$

All probabilities remain in a fixed compact subset of $(0, 1)$. Moreover,

$$r_n'(x) = \frac{1}{20} + \frac{1}{100}\cos(nx) \in [0.04, 0.06], \tag{28}$$

so every map is a regular immersion with a uniform rank bound. Uniform interiority and Taylor's theorem give $\sup_x D_{\mathrm{KL}}(p_* \| p_n) = O(n^{-2})$. The Bernoulli pullback metric is

$$g(x) = \frac{r'(x)^2}{r(x)(1 - r(x))}\,\mathrm{d}x^2. \tag{29}$$

At $x = 0$, $r_n(0) = r(0) = 1/3$, but $r_n'(0) = 3/50$ and $r'(0) = 1/20$. Hence

$$g_n(0) = \frac{81}{5000}\,\mathrm{d}x^2 \neq \frac{9}{800}\,\mathrm{d}x^2 = g_*(0). \tag{30}$$

□

This example rules out an unconditional theorem of the form "cross-entropy convergence implies Fisher convergence." The oscillations have bounded first derivative but unbounded higher regularity, anticipating the Sobolev condition in Section 7.

There is a second nonidentifiability. For any hidden diffeomorphism ψ,

$$H' = \psi \circ H, \qquad F' = F \circ \psi^{-1} \tag{31}$$

produces exactly the same predictions while changing ambient Euclidean coordinates. Pullback Fisher geometry handles this gauge covariantly, but loss alone cannot identify a privileged hidden Euclidean metric.

6 Stability of Fisher geometry and transport

We now quantify how derivative-level predictive agreement propagates to geometry. We state the result on a common Euclidean chart. For two latent manifolds related by a comparison diffeomorphism Φ, replace the second map by $\widetilde{p} = F_B \circ \Phi$.

Let $K \Subset U \subset \mathbb{R}^m$ and let $p, \widetilde{p} : U \to \Delta_N^\circ$. For an integer k, use the componentwise norm

$$\|p\|_{C^k(K)} = \max_{a, |\beta| \le k} \sup_{x \in K} |\partial^\beta p_a(x)|. \tag{32}$$

Assumption 6.1 (Uniform regularity)**.** For constants $\eta, \lambda, B_2 > 0$,

$$p_a(x), \widetilde{p}_a(x) \ge \eta, \tag{33}$$

$$\|p\|_{C^2(K)} + \|\widetilde{p}\|_{C^2(K)} \le B_2, \tag{34}$$

$$\lambda_{\min}(g_p(x)), \lambda_{\min}(g_{\widetilde{p}}(x)) \ge \lambda \tag{35}$$

for every a and $x \in K$.

The probability floor excludes the singular simplex boundary. The spectral floor excludes decoder-null directions after quotienting and prevents inverse-metric instability.

Theorem 6.2 (C^2 stability of α-transport)**.** *Under Assumption 6.1, fix $A > 0$. There is a constant*

$$C_\Gamma = C_\Gamma(m, N, A, \eta^{-1}, \lambda^{-1}, B_2) \tag{36}$$

such that, uniformly for $|\alpha| \leq A$,

$$\begin{aligned} &\|g_p - g_{\widetilde{p}}\|_{C^1(K)} + \|C_p - C_{\widetilde{p}}\|_{C^0(K)} \\ &\quad + \|\Gamma_p^{(\alpha)} - \Gamma_{\widetilde{p}}^{(\alpha)}\|_{C^0(K)} \leq C_\Gamma \|p - \widetilde{p}\|_{C^2(K)}. \end{aligned} \tag{37}$$

Let $\gamma : [0, 1] \to K$ be piecewise C^1, with Euclidean length L_γ. If

$$M_\Gamma = \max\{\|\Gamma_p^{(\alpha)}\|_{C^0}, \|\Gamma_{\widetilde{p}}^{(\alpha)}\|_{C^0}\}, \tag{38}$$

where the norm is the induced bilinear operator norm, then

$$\|P_\gamma^{p,\alpha} - P_\gamma^{\widetilde{p},\alpha}\|_{\mathrm{op}} \leq L_\gamma e^{M_\Gamma L_\gamma} C_\Gamma \|p - \widetilde{p}\|_{C^2(K)}. \tag{39}$$

Proof. Equations (7) and (8) are finite sums of rational functions of p and its first derivatives. Their denominators are controlled by η. Differentiating (7) once uses at most second derivatives, giving the first two terms of (37) by the mean-value theorem on the compact regularity class.

For completeness, the lower-index α-coefficients have the direct probability-coordinate expression

$$\Gamma_{ij,k}^{(\alpha)} = \sum_a \left[\frac{\partial_{ij} p_a \, \partial_k p_a}{p_a} - \frac{1 + \alpha}{2} \frac{\partial_i p_a \, \partial_j p_a \, \partial_k p_a}{p_a^2} \right]. \tag{40}$$

Thus their difference is $O(\|p - \widetilde{p}\|_{C^2})$. Raising the final index requires inverse metrics. The identity

$$g_p^{-1} - g_{\widetilde{p}}^{-1} = g_p^{-1}(g_{\widetilde{p}} - g_p) g_{\widetilde{p}}^{-1} \tag{41}$$

and the spectral floor λ give inverse-metric stability, proving (37).

Along γ, transport solves the linear ODE

$$\dot{V}(t) = -\Gamma^{(\alpha)}(\gamma(t))[\dot{\gamma}(t)] V(t). \tag{42}$$

Duhamel's formula bounds the difference of the two fundamental solutions by the integral of the coefficient defect multiplied by their growth factors. Gronwall's inequality yields (39). ☐

Corollary 6.3 (Cross-model transport commutator)**.** *Let $F_A : M_A \to \Delta_N^\circ$, $F_B : M_B \to \Delta_N^\circ$, and let $\Phi : M_A \to M_B$ be a C^2 local diffeomorphism. Suppose F_A and $F_B \circ \Phi$ satisfy Assumption 6.1 and*

$$\|F_A - F_B \circ \Phi\|_{C^2(K)} \leq \varepsilon. \tag{43}$$

Then

$$\begin{aligned} &\big\| \mathrm{d}\Phi_{\gamma(1)} P_\gamma^{A,\alpha} - P_{\Phi\circ\gamma}^{B,\alpha} \mathrm{d}\Phi_{\gamma(0)} \big\|_{\mathrm{op}} \\ &\qquad \leq L_\gamma e^{M_\Gamma L_\gamma} C_\Gamma \varepsilon, \end{aligned} \tag{44}$$

after using the aligned tangent norms. At $\varepsilon = 0$, this reduces to Theorem 4.1.

Proof. The connection induced by $F_B \circ \Phi$ is the pullback of the connection induced by F_B. Its transport is conjugated by $\mathrm{d}\Phi$. Apply Theorem 6.2 on M_A. ☐

Theorem 6.4 (C^3 curvature stability). *Assume the hypotheses of Theorem 6.2, replace the C^2 bound by*

$$\|p\|_{C^3(K)} + \|\widetilde{p}\|_{C^3(K)} \le B_3, \tag{45}$$

and retain the probability and spectral floors. Then

$$\|R^{(\alpha)}_p - R^{(\alpha)}_{\widetilde{p}}\|_{C^0(K)} \le C_R \|p - \widetilde{p}\|_{C^3(K)}, \tag{46}$$

where C_R depends on $m, N, A, \eta^{-1}, \lambda^{-1}, B_3$. Levi-Civita sectional curvatures also converge on corresponding two-planes whose Gram determinants are uniformly bounded below.

Proof. Differentiating the argument for (37) gives C^1 control of the Christoffel symbols. Insert it into

$$R^{\ell}{}_{ijk} = \partial_i \Gamma^{\ell}{}_{jk} - \partial_j \Gamma^{\ell}{}_{ik} + \Gamma^{\ell}{}_{ir}\Gamma^{r}{}_{jk} - \Gamma^{\ell}{}_{jr}\Gamma^{r}{}_{ik}. \tag{47}$$

Sectional curvature is a quotient. The same estimates control its metric numerator, while the uniform Gram bound controls its denominator. □

Remark 6.5 (Why the assumptions matter). The constants diverge near $p_a = 0$ because the Fisher tensors contain p_a^{-1} and p_a^{-2}. They also diverge near rank loss because raising indices requires g^{-1}. These are geometric instabilities, not proof artifacts. In large-vocabulary models, a useful extension would replace a uniform tokenwise probability floor by moment or coarse-graining conditions with vocabulary-independent constants.

7 From cross-entropy to transport under Sobolev regularity

Theorem 6.2 assumes derivative-level agreement. We now give sufficient conditions under which population KL convergence implies it.

Let Ω be a compact m-dimensional smooth manifold or a bounded Sobolev-extension domain. Let μ have density ρ with respect to volume, satisfying $\rho \ge \rho_0 > 0$. Define

$$\mathcal{K}(p, \widetilde{p}) = \int_\Omega D_{\mathrm{KL}}\big(p(x)\|\widetilde{p}(x)\big)\, \mathrm{d}\mu(x). \tag{48}$$

Theorem 7.1 (Risk-to-transport convergence). *Let $p, \widetilde{p} : \Omega \to \Delta_N^\circ$ satisfy the probability and spectral floors of Assumption 6.1. Suppose*

$$p, \widetilde{p} \in H^s(\Omega; \mathbb{R}^N), \qquad s > 2 + \frac{m}{2}, \tag{49}$$

and

$$\|p\|_{H^s} + \|\widetilde{p}\|_{H^s} \le B. \tag{50}$$

For every r with

$$2 + \frac{m}{2} < r < s, \tag{51}$$

there is a constant C such that, for every bounded-length path γ,

$$\|P^{p,\alpha}_\gamma - P^{\widetilde{p},\alpha}_\gamma\|_{\mathrm{op}} \le C B^{r/s} \mathcal{K}(p, \widetilde{p})^{\frac{s-r}{2s}}. \tag{52}$$

Equivalently, the transport error is $O(\mathcal{K}^\beta)$ for every

$$0 < \beta < \frac{s - 2 - m/2}{2s}. \tag{53}$$

Proof. Pinsker's inequality and $\rho \geq \rho_0$ give

$$\begin{aligned} \|p - \widetilde{p}\|^2_{L^2(\mathrm{dvol})} &\leq \rho_0^{-1} \int_\Omega \|p - \widetilde{p}\|_1^2 \,\mathrm{d}\mu \\ &\leq 2\rho_0^{-1} \mathcal{K}(p, \widetilde{p}). \end{aligned} \tag{54}$$

Sobolev interpolation [1] yields

$$\|p - \widetilde{p}\|_{H^r} \leq C \|p - \widetilde{p}\|_{L^2}^{1-r/s} \|p - \widetilde{p}\|_{H^s}^{r/s}, \tag{55}$$

and therefore

$$\|p - \widetilde{p}\|_{H^r} \leq C B^{r/s} \mathcal{K}(p, \widetilde{p})^{(s-r)/(2s)}. \tag{56}$$

Since $r > 2 + m/2$, Sobolev embedding gives $H^r \hookrightarrow C^2$. Apply Theorem 6.2. ☐

Corollary 7.2 (Two-model geometric convergence)**.** *Suppose two model sequences $p_{A,n}, p_{B,n}$ on a shared intervention chart converge in population cross-entropy to the same conditional map p_*, and all three maps satisfy uniform versions of the hypotheses of Theorem 7.1. Then, for every admissible β,*

$$\|P_\gamma^{A,n,\alpha} - P_\gamma^{B,n,\alpha}\|_{\mathrm{op}} \leq C(\epsilon_{A,n}^\beta + \epsilon_{B,n}^\beta), \tag{57}$$

where $\epsilon_{i,n} = \mathbb{E} D_{\mathrm{KL}}(p_* \| p_{i,n})$. With a cross-model alignment Φ_n, the same statement holds for the transport commutator in (44) after replacing $p_{B,n}$ by $p_{B,n} \circ \Phi_n$.

If $s > 3 + m/2$, the same argument combined with Theorem 6.4 gives curvature convergence for every

$$0 < \beta < \frac{s - 3 - m/2}{2s}. \tag{58}$$

The uniform Sobolev hypothesis excludes the high-frequency behavior in Proposition 5.1. For language models it should be tested on low-dimensional continuous intervention charts, such as soft-prompt paths or controlled activation subspaces, rather than assumed on the discrete set of tokenized contexts.

8 The semantic-sufficiency obstruction

Even perfect agreement of predictive geometry does not guarantee that a semantic operation is a field on that geometry. We now separate this representational obstruction from connection mismatch.

Let $F : \mathcal{H} \to P$ be a regular predictive quotient, let H be a random hidden state with law μ, and let $X_T(H) \in T_H\mathcal{H}$ describe a semantic operation T. Its observable infinitesimal predictive effect is

$$Z_T(H) = \mathrm{d}F_H X_T(H) \in T_{F(H)}P. \tag{59}$$

Assume square integrability in the Fisher metric. Since P is embedded in the simplex, tangent-valued conditional expectations may be taken fiberwise in the ambient zero-sum space; conditioning on $F(H) = p$ fixes both T_pP and g_p.

Theorem 8.1 (Semantic conditional-variance obstruction)**.** *For a measurable, square-integrable field $Y : P \to TP$, define*

$$\mathcal{E}_T(Y) = \mathbb{E}\|Z_T(H) - Y(F(H))\|^2_{g_{F(H)}}. \tag{60}$$

Let

$$\overline{Z}_T(p) = \mathbb{E}[Z_T(H) \mid F(H) = p]. \tag{61}$$

Then

$$\begin{aligned}\mathcal{E}_T(Y) = \mathbb{E}\|Z_T - \overline{Z}_T(F(H))\|_g^2 \\ + \mathbb{E}\|\overline{Z}_T(F(H)) - Y(F(H))\|_g^2.\end{aligned} \tag{62}$$

Consequently,

$$\inf_Y \mathcal{E}_T(Y) = \mathbb{E}\,\mathrm{Var}_g(Z_T \mid F). \tag{63}$$

The semantic effect descends almost surely to a predictive field if and only if this conditional variance is zero. Smooth descent additionally requires a smooth version of $\overline{Z}_T$.

Proof. Square-integrable tangent-valued random variables form a Hilbert direct integral with inner product

$$\langle U, V\rangle = \mathbb{E}[g_{F(H)}(U, V)]. \tag{64}$$

The fields measurable with respect to $\sigma(F)$ form a closed subspace. Conditional expectation is the orthogonal projection onto that subspace. Equivalently, decompose

$$\begin{aligned}Z_T - Y(F) = \big(Z_T - \overline{Z}_T(F)\big) \\ + \big(\overline{Z}_T(F) - Y(F)\big).\end{aligned} \tag{65}$$

The two terms are orthogonal after conditioning on F, which proves (62) and (63). ☐

Corollary 8.2 (Connection-restricted decomposition)**.** *Let $\mathcal{C}_\alpha$ be any closed class of predictive fields selected by an α-connection, for example fields satisfying $\nabla^{(\alpha)}Y = 0$ on a specified simply connected chart or fields with a bounded covariant-derivative residual. Then*

$$\begin{aligned}\inf_{Y \in \mathcal{C}_\alpha} \mathcal{E}_T(Y) = \underbrace{\mathbb{E}\,\mathrm{Var}_g(Z_T \mid F)}_{\textit{predictive insufficiency}} \\ + \underbrace{\inf_{Y \in \mathcal{C}_\alpha} \mathbb{E}\|\overline{Z}_T - Y\|_g^2}_{\textit{connection mismatch}}.\end{aligned} \tag{66}$$

Corollary 8.3 (Cross-model semantic transfer)**.** *Let P_A, P_B be predictive quotients related by a Fisher isometry $\Phi : P_A \to P_B$ on paired samples. Write $Q_i = F_i(H_i)$ for the random predictive state of model i. Define*

$$Y_{A\to B}(p_B) = \mathrm{d}\Phi_{\Phi^{-1}(p_B)}\overline{Z}_A(\Phi^{-1}(p_B)). \tag{67}$$

Then

$$\begin{aligned}\mathbb{E}\|Z_B - Y_{A\to B}(Q_B)\|_{g_B}^2 = \underbrace{\mathbb{E}\|Z_B - \overline{Z}_B(Q_B)\|_{g_B}^2}_{\textit{model-B quotient obstruction}} \\ + \underbrace{\mathbb{E}\|\overline{Z}_B - \mathrm{d}\Phi\,\overline{Z}_A\|_{g_B}^2}_{\textit{cross-model field mismatch}}.\end{aligned} \tag{68}$$

No choice of α can reduce the first term of (66). A strictly positive population conditional Fisher variance therefore proves that no predictive-state-only field, and hence no choice of α-transport on that quotient, can represent the specified infinitesimal effect exactly.

9 Operational tests and training implications

The theorems suggest a hierarchy of increasingly strong cross-model agreement tests:

$$\begin{array}{lll} \text{Level 0} & : & \text{output KL agreement,} \\ \text{Level 1} & : & \text{Fisher metric agreement,} \\ \text{Level 2} & : & \alpha\text{-connection and transport agreement,} \\ \text{Level 3} & : & \text{semantic-field equivariance,} \\ \text{Level 4} & : & \text{curvature and holonomy agreement.} \end{array} \tag{69}$$

Agreement at one level does not imply the next without the regularity and sufficiency conditions identified above.

9.1 Transport commutation score

For an estimated alignment Φ and sampled paths and tangent vectors, define

$$\mathcal{D}_{\mathrm{PT}}^{(\alpha)}(A, B; \Phi) = \mathbb{E}_{\gamma,v} \frac{\|\mathrm{d}\Phi_{\gamma(1)} P_\gamma^{A,\alpha} v - P_{\Phi\gamma}^{B,\alpha} \mathrm{d}\Phi_{\gamma(0)} v\|}{\|v\|}. \tag{70}$$

Theorem 4.1 makes this score zero under exact predictive equivalence, while Corollary 6.3 bounds it under approximate equivalence. A practical study should test whether (70) predicts held-out steering transfer or task generalization beyond output KL and representation-only similarities such as CKA.

9.2 Selecting the connection

At a linear softmax head, $\alpha = 1$ is ordinary vector reuse, $\alpha = 0$ is Fisher-metric-compatible Levi-Civita transport, and $\alpha = -1$ is mixture transport. A predictive semantic field Y_T can be evaluated through

$$\mathcal{R}_T(\alpha) = \mathbb{E}\|\nabla^{(\alpha)} Y_T\|_g^2. \tag{71}$$

The minimizer is interpreted relative to the tested field and chart:

- a minimum at the exponential connection means that the tested field is closest to exponential-parallel; at the affine softmax head this is constant natural-coordinate vector reuse;
- a minimum at the Levi-Civita connection means that the tested field is closest to the metric-compatible transport;
- a minimum at the mixture or an intermediate α identifies the corresponding affine connection as the closest member of the tested family; intermediate α-connections need not be flat;
- positive population conditional variance proves predictive insufficiency independently of the minimizing connection;
- small estimated conditional variance together with large residuals for every tested α motivates an operation-specific connection or a nonparallel field model.

9.3 Training regimes

The risk-to-transport theorem indicates why derivative regularity matters. A causal training study can compare cross-entropy-only models with matched-loss models trained using

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \widehat{\text{Var}}_g(\mathrm{d}FX_T \mid F) + \mu \|\nabla^{(\alpha)} \overline{Z}_T\|_g^2 + \nu \mathcal{R}_{\text{smooth}}. \tag{72}$$

The first regularizer encourages the operation to descend to the predictive quotient. The second selects connection-relative consistency. The final term is intended to enforce the Sobolev-type regularity required by Theorem 7.1; applying the theorem requires verifying the corresponding uniform bound. The empirical protocol treats behavioral improvement as the primary outcome and the geometric regularizers as diagnostic quantities.

A no-GPU pilot can use frozen small models and one- or two-dimensional soft-prompt charts. Finite differences estimate F, ∂F, and $\partial^2 F$; the model's output head provides exact vocabulary probabilities. The main comparisons are output KL, metric defect, the transport score (70), semantic conditional variance, and held-out intervention transfer.

10 Limitations

Continuous charts. Tokenized contexts are discrete. Derivatives and connections require an explicit continuous chart, such as soft prompts, activation interventions, or a justified local manifold model.

Boundary and rank. Softmax probabilities are positive but can be extremely small. Uniform tokenwise floors are mathematically clean and practically strong. The Fisher metric can also be severely ill-conditioned. Quotienting, regularization, coarse-graining, or horizontal-subspace analysis is necessary near rank loss.

Different tokenizers. The exact theorem assumes a common outcome space. A tokenizer map generally acts as a Markov morphism. Fisher geometry is preserved only under a statistically sufficient or reversible identification; otherwise it contracts [5].

Population versus empirical loss. Equation (24) concerns population risk, realizability, and global optimization. Finite empirical loss introduces estimation and optimization errors not treated here.

Alignment is not identified automatically. The stability theorem evaluates a proposed Φ. Constructing a statistically identifiable alignment from finite observations, especially across different latent dimensions, is a separate problem. One may instead work on a shared low-dimensional intervention chart.

Connection dependence. Parallel transport compares tangent vectors at different points relative to a selected connection. At the final affine softmax head, ambient addition coincides with exponential-affine transport, while Levi-Civita and mixture transport preserve different structures. The theory therefore evaluates connection choice separately across operations, layers, and tasks.

11 Conclusion

Idealized pretraining makes convergence of conditional output distributions natural because excess cross-entropy is expected KL divergence. It does not by itself make convergence of Fisher metrics, affine connections, or curvature natural. Geometry depends on derivatives, and a regular oscillatory counterexample separates these notions.

Under explicit smoothness and identifiability assumptions, the stronger conclusion becomes provable. Predictive maps that converge in C^2 have convergent Amari α-connections and parallel transports; C^3 convergence controls curvature. Integrated KL plus a uniform Sobolev bound supplies the missing derivative convergence and yields a quantitative loss-to-transport theorem. The semantic conditional-variance obstruction then determines whether the predictive quotient is rich enough to support the operation at all.

If aligned predictive maps have vanishing population KL risk together with the uniform Sobolev, boundary, and rank bounds of Theorem 7.1, their α-transport commutators vanish. Semantic-field equivariance is a separate condition: it requires small within-model quotient variance and small cross-model conditional-mean-field mismatch. These quantities localize deviations from transfer to inadequate regularity, predictive insufficiency, field mismatch, or connection mismatch.

Code and manuscript availability

The manuscript source, model-free predictive-geometry implementation, CPU tests, and experimental protocols are available at

https://github.com/guybass/probalistic-latent-spaces.

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